Immunodeficiency 127 (IMD127) — Comprehensive Disease Characteristics Report

Prepared: 2026-09-02 · Autonomous literature/database synthesis **Primary source:** Arias, Neehus, Ogishi et al. "Tuberculosis in otherwise healthy adults with inherited TNF deficiency." *Nature* 633:417–425 (2024). ;

P 39198650

DOI 10.1038/s41586-024-07866-3. OMIM #**620977**.

> **Evidence caveat.** IMD127 is an ultra-rare, newly delineated Mendelian disorder described in **a single consanguineous Colombian family (2 affected first cousins)**. Nearly all disease-specific clinical facts derive from this one primary report. Mechanistic corroboration comes from in vitro/iPSC studies, mouse models, and the large pharmacovigilance literature on anti-TNF-associated tuberculosis. Where a statement is inference or extrapolation (e.g., standard TB therapy, epidemiology of TB generally), it is flagged.

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1. Disease Information

Overview. Immunodeficiency-127 is an **autosomal recessive inborn error of immunity** characterized by **selective susceptibility to pulmonary tuberculosis (TB)** in otherwise healthy individuals. It is the **first described human inherited TNF (tumor necrosis factor) deficiency**. Affected individuals tolerate BCG vaccination normally, develop recurrent pulmonary *Mycobacterium tuberculosis* infection beginning in the late teens, and have no other consistent infectious or immunologic abnormality (Arias et al. 2024,).

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Key identifiers. - **OMIM:** #620977 (IMMUNODEFICIENCY 127; IMD127); causative gene TNF #191160. - **MONDO: MONDO:0975832** ("immunodeficiency 127") — confirmed via EBI OLS; maps to OMIM:620977. - **Orphanet:** No specific ORPHA code identified for this newly described entity; conceptually within "Mendelian susceptibility to mycobacterial diseases" (ORPHA:319583) and rare primary immunodeficiency groupings. - **ICD-11:** Best fit **4A00.0** (primary immunodeficiencies) / predisposition to infection; the clinical infection codes as tuberculosis (ICD-11 **1B1**; ICD-10 **A15–A19**). - **ICD-10:** D84.9 (immunodeficiency, unspecified) for the trait; A15.- (respiratory TB, bacteriologically confirmed) for disease episodes. - **MeSH:** No dedicated descriptor; relevant terms: "Tuberculosis, Pulmonary" (D014397), "Tumor Necrosis Factor-alpha" (D014409), "Immunologic Deficiency Syndromes" (D007153).

Synonyms / alternative names: Inherited TNF deficiency; **Autosomal recessive complete TNF deficiency** (OMIM's descriptor); Human TNF deficiency; TNF-deficiency Mendelian susceptibility to tuberculosis. OMIM notes the disorder is characterized by ROS-deficient alveolar macrophages, TB onset late in the second decade with common relapse, and infections that respond to medication.

Data provenance: Disease-level knowledge is derived from **individual patient data** (deep immunophenotyping of 2 related patients plus family segregation), complemented by **aggregated/experimental** resources (iPSC models, mouse KO literature, anti-TNF pharmacovigilance). Not an EHR-derived aggregate.

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2. Etiology

Primary cause — genetic. Biallelic (homozygous) **loss-of-function variant in TNF** (6p21.33). The reported allele is a private frameshift (Section 4). Disease requires infectious exposure to *M. tuberculosis*: the genetic lesion creates susceptibility, and the environmental trigger (Mtb) precipitates disease → a clear **gene** × **environment (infection) interaction**.

Genetic risk factors. - *Causal:* homozygous TNF null (`c.190_191ins20`, p.Pro64LeufsTer13). - *Consanguinity* is a major enabling factor (autosomal recessive, private variant; patient homozygosity 1.14% and 2.1%). (P 39198650)

- *Related monogenic TB loci* (not modifiers of IMD127 per se, but the differential): TYK2 (incl.

common P1104A), IL12RB1, IL12B, IL23R, IFNGR1/2, STAT1, ISG15 (P 32055999 P 38025345).

Environmental risk factor (obligate trigger): exposure to/infection with *Mycobacterium tuberculosis* (NCBI Taxon 1773). General TB risk amplifiers (crowding, HIV, malnutrition, high-burden geography) plausibly apply but are unstudied in IMD127.

Protective factors. Not established. By inference, avoidance of Mtb exposure and latent-TB prophylaxis would be protective. Heterozygous carriers are clinically unaffected (recessive). No protective alleles described.

Gene–environment interaction: TNF-null genotype is clinically silent until Mtb challenge; conversely Mtb is controlled in TNF-sufficient hosts. This mirrors the acquired phenocopy in which anti-TNF biologics unmask latent TB (Section 12; P 29459143).

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3. Phenotypes

Because n=2, "frequencies" below reflect the reported family; interpret cautiously.

Phenotype	Type	HPO term	Onset	Severity/ course	Frequency (this family)
Recurrent pulmonary tuberculosis	Clinical sign / infectious	**HP:0032262** (Pulmonary tuberculosis); parent HP:0032251 Tuberculosis; HP:0002205 Recurrent respiratory infections	Late teens (18, 19 y)	Recurrent/ relapsing but treatment-responsive	2/2 (100%)
Increased susceptibility to mycobacterial infection	Lab/ immunologic predisposition	**HP:0004385** related; HP:0002718 (Recurrent bacterial infections)	Adolescent–adult	Selective to Mtb	2/2
Severe *Listeria monocytogenes* infection (during pregnancy)	Clinical sign / infectious	HP:0031386 (Listeria) / HP:0002718	Adult	Severe, single episode	1/2
Normal BCG-vaccine tolerance (no BCG-osis)	Absence of expected phenotype	(negative finding)	—	—	2/2
Normal blood leukocyte subsets	Lab (normal)	—	—	Stable	2/2
Normal clinical/ biological inflammatory responses	Lab (normal)	—	—	Normal	2/2

Notable negatives (diagnostically important): No disseminated/extrapulmonary mycobacterial disease, no adverse vaccine reactions, no broad susceptibility to viral/fungal/pyogenic organisms, no autoimmunity, no developmental phenotype.

Quality-of-life impact: Driven by recurrent TB episodes (cough, fever, weight loss, treatment burden of multi-month antibiotics, relapse anxiety). Between episodes, patients are functionally healthy. No formal QoL instruments (EQ-5D/SF-36) reported.

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4. Genetic / Molecular Information

Causal gene: TNF (tumor necrosis factor / TNF-alpha). - HGNC: **11892**; NCBI Gene: **7124**; Ensembl: **ENSG00000232810**; UniProt: **P01375**; OMIM gene: **191160**. - Locus: **chromosome 6p21.33**, within the **MHC class III region** (between LTA and LTB/HLA-B).

Pathogenic variant (OMIM 191160.0007): - cDNA: `c.190_191ins20` (20-nt insertion), exon 2. - **Protein:** `p.Pro64LeufsTer13` (P64Lfs*13) — frameshift beginning at codon 64 with a premature stop 13 residues downstream. - Type/class: frameshift insertion → premature termination codon (predicted NMD and/or truncated non-functional protein). - Zygosity/origin: homozygous germline; private to the family; absent in extended relatives and population databases (MAF <0.01 filter; not observed in gnomAD). - Functional consequence: complete loss of function. HEK293 transduced with mutant TNF failed to induce NF-κB-dependent transcription; patient cells showed absent TNF expression, no secreted TNF, and no TNF induction to LPS/BCG/Listeria/IFNγ; rescued by wildtype TNF (P 39198650). - ACMG/AMP classification: Pathogenic* — null variant in a gene with established LOF mechanism (PVS1), co-segregation, absence in controls (PM2), functional damage (PS3).

Modifier genes: none identified (single family).

Epigenetic information: None reported for IMD127. (TNF expression is normally under complex transcriptional/epigenetic control at the MHC-III locus, but no disease-specific epigenetic change is described.)

Chromosomal abnormalities: None; point-level insertion, no cytogenetic lesion.

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5. Environmental Information

- **Infectious agent (essential):** *Mycobacterium tuberculosis* (NCBI Taxon **1773**) — the disease-defining pathogen. *Listeria monocytogenes* (NCBI Taxon **1639**) caused one severe episode.
- **Weakly virulent mycobacteria tolerated:** *M. bovis* BCG (vaccine) was handled normally — a key contrast to IFN-γ-circuit defects.
- **Environmental/lifestyle/toxic factors:** none specifically implicated; general TB determinants (exposure intensity, HIV, crowding) presumably relevant but unstudied.

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6. Mechanism / Pathophysiology

Causal chain (initiating lesion → clinical manifestation)

1. **Homozygous TNF frameshift `c.190_191ins20` (p.P64Lfs*13)** *leads to* absence of functional TNF protein (no intracellular, transmembrane, or secreted TNF). *(Demonstrated: absent expression/secretion.)*
2. Absence of TNF *results in* loss of **autocrine/paracrine TNF→TNFR1 (TNFRSF1A) signaling** in mononuclear phagocytes. *(Demonstrated: no NF-κB induction; TNFR1-KO iPSC macrophages phenocopy.)*
3. Loss of TNF–TNFR1 signaling in **GM-CSF-matured monocyte-derived macrophages and alveolar-macrophage-like cells** *leads to* failure of the **NADPH-oxidase respiratory burst** (deficient reactive oxygen species production). *(Demonstrated in patient cells, TNF- and TNFR1-deficient iPSC macrophages, and TNF-blocker-treated control/lung macrophages.)*
4. Impaired respiratory burst *results in* **defective ROS-dependent intracellular killing of ingested *M. tuberculosis*** within lung macrophages. *(Inferred from respiratory-burst defect + rescue by TNF.)*
5. Defective mycobacterial killing in the alveolar compartment *leads to* **uncontrolled Mtb replication and recurrent pulmonary tuberculosis**. *(Observed clinical outcome.)*

Branch point / redundancy: Because TNF is **redundant** for (a) IFN-γ production (patients' IFN-γ and downstream cytokines were intact), (b) leukocyte development, (c) systemic inflammation, and (d) control of BCG and most other microbes, the phenotype is **narrowly restricted to Mtb in the lung**. This contrasts with (i) complete phagocyte respiratory-burst deficiency (chronic granulomatous disease), which causes multi-organism infection, and (ii)

IFN-γ-circuit MSMD defects, which cause BCG-osis + environmental mycobacterial disease (P 39198650 P 32055999).

Mechanistic detail by category

- **Molecular pathways:** TNF–TNFR1 signaling → NF-κB activation (canonical); coupling to **NADPH oxidase (respiratory burst)**. KEGG **hsa04668** (TNF signaling), Reactome **R-HSA-75893** (TNF signaling), **R-HSA-1222556** (NF-κB). ROS generation ~ Reactome R-HSA-1222556 / phagosome pathways.
- **Cellular processes:** phagocytosis and **oxidative microbial killing** in macrophages; innate immune activation. GO: **GO:0045087** innate immune response; **GO:0006801** superoxide metabolic process; **GO:0045730** respiratory burst; **GO:0034612** response to TNF; **GO:0032496** response to LPS; **GO:0050832** defense response to fungus (n/a); **GO:0071356** cellular response to TNF.
- **Protein dysfunction:** TNF is a type II transmembrane homotrimeric cytokine cleaved by **TACE/ADAM17** to soluble TNF; the frameshift abolishes the mature TNF homology domain → loss of function (no misfolding/aggregation/gain-of-function).
- **Immune involvement:** isolated innate-immune effector defect (macrophage killing); adaptive immunity, granuloma-relevant cytokines (IFN-γ) intact clinically.
- **Tissue damage:** driven by unchecked Mtb infection (caseating granulomatous inflammation of lung) rather than a primary tissue-toxic mechanism.
- **Biochemical abnormality:** functional deficiency of the macrophage **respiratory burst / NADPH oxidase output** secondary to absent TNF signaling (not a primary CYBB/NCF defect).
- **Molecular profiling:** iPSC-derived macrophage models (TNF-KO, TNFR1-KO) used for functional dissection; no patient transcriptomic/proteomic/metabolomic datasets published. Respiratory-burst assays are the key functional readout.

Cell types (CL): CL:0000583 alveolar macrophage; **CL:0000235** macrophage; **CL:0001054** CD14+ monocyte; **CL:0000576** monocyte. **Biological process (GO):** GO:0045730 (respiratory burst), GO:0071356 (cellular response to TNF).

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7. Anatomical Structures Affected

- **Primary organ: lung** (UBERON:0002048); specifically alveolar/pulmonary parenchyma (UBERON:0002299 alveolus). Disease is **pulmonary and typically bilateral** as per TB.
- **Body system: respiratory system** (UBERON:0001004); **immune system** (UBERON:0002405) as the affected functional system.
- **Secondary involvement:** one patient had disseminated *Listeria* (systemic). Classic TB complications (pleura, mediastinal lymph nodes) possible but not emphasized.
- **Tissue/cell level:** mononuclear phagocytes — **alveolar macrophages (CL:0000583)** and monocyte-derived macrophages — are the pathogenically central cells; epithelial/lymphoid compartments intact.
- **Subcellular (GO cellular component):** phagosome (**GO:0045335**), plasma membrane (**GO:0005886**), NADPH oxidase complex (**GO:0043020**), extracellular space (**GO:0005615**) for secreted TNF.
- **Lateralization:** pulmonary TB is generally **bilateral/multifocal**; not a lateralized trait.

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8. Temporal Development

- **Onset: adolescent/young-adult** — first TB episodes at ages **18 and 19**. No neonatal/childhood infections despite BCG exposure. Onset pattern: **subacute/chronic** (typical TB).
- **Progression: relapsing/recurrent** pulmonary TB; individual episodes are treatment-responsive; disease course over the reported follow-up shows **recurrence rather than relentless progression**.
- **Duration:** the underlying immunodeficiency is **lifelong**; clinical disease is **episodic**, contingent on Mtb exposure/reactivation.
- **Remission: treatment-induced** remission of each episode with anti-TB chemotherapy; relapse may occur.
- **Critical period / intervention window:** post-exposure and early infection — latent-TB detection and prophylaxis represent the key opportunity to prevent active disease.

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9. Inheritance and Population

- **Inheritance: Autosomal recessive** (biallelic TNF LOF). Heterozygotes unaffected.
- **Epidemiology: Ultra-rare** — to date **1 family, 2 patients** worldwide (P 39198650). No prevalence/incidence estimate; effectively <1 per 10^6 (private variant). Broader context: only ~5–10% of Mtb-infected individuals develop TB, and monogenic causes account for a small fraction (e.g., TYK2 P1104A ~1% of TB in Europeans; P 32055999).
- **Penetrance:** appears **high upon Mtb exposure** but is **environment-dependent** (requires infection); true penetrance unknown with n=2.
- **Expressivity:** narrow, consistent (isolated pulmonary TB) in the two cases.
- **Anticipation / mosaicism:** not applicable / not reported.
- **Founder effect:** none — variant is **private** to this family.
- **Consanguinity:** central — multigenerational consanguineous pedigree; homozygosity mapping/linkage used for discovery.
- **Carrier frequency:** effectively 0 in populations (variant absent from gnomAD).
- **Population/geography:** single **Colombian** family; no ethnic predilection can be inferred. Sex: both cases female-inclusive (one episode occurred in pregnancy); no sex-ratio inference possible.

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10. Diagnostics

Clinical/laboratory workup: - **Confirm TB:** sputum smear microscopy for acid-fast bacilli, mycobacterial culture, NAAT (e.g., Xpert MTB/RIF), chest imaging (X-ray/CT showing pulmonary infiltrates/cavitation). (Standard TB diagnostics; RadLex/LOINC applicable.) - **Immunologic screen (typically normal in IMD127):** normal CBC/leukocyte subsets, immunoglobulins, and IFN-γ responses — normality helps distinguish from other IEIs. - **Functional immunology (research/specialist): absent TNF production** by stimulated whole blood/monocytes (LPS, BCG, IFN-γ) and **impaired macrophage respiratory burst** — the

pathognomonic functional signature (P 39198650).
- **Biomarker:** undetectable serum/secreted TNF after stimulation; the defect is the biomarker.

Genetic testing (definitive): - **WES or WGS** with homozygosity mapping is how the diagnosis was made; **single-gene TNF sequencing** confirms once suspected. - **Gene panels:** include TNF alongside MSMD/IEI-to-TB genes (TYK2, IL12RB1, IL12B, IL23R, IFNGR1, IFNGR2, STAT1, ISG15, CYBB/CGD genes). - CMA/karyotype/FISH/mtDNA/repeat testing: **not applicable** (point insertion).

Clinical criteria / differential diagnosis: No formal diagnostic criteria (novel disease). Diagnose when a young adult has **recurrent isolated pulmonary TB, normal BCG tolerance, no other infections, normal IFN-γ axis, and biallelic TNF LOF**. **Differential:** TYK2 deficiency (complete or P1104A homozygosity), IL-12Rβ1 deficiency, IFN-γR/STAT1 defects (usually BCG-osis + broader mycobacterial disease), chronic granulomatous disease (broad infection spectrum), HIV and acquired anti-TNF exposure (P 32055999 P 38025345 P 39198650).

Screening: cascade genetic testing of relatives; latent-TB screening (IGRA/TST) in gene-positive individuals.

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11. Outcome / Prognosis

- **Survival/mortality:** No deaths reported among the 2 patients; **most TB episodes were successfully treated** (P 39198650). Untreated pulmonary TB carries substantial mortality generally, but with therapy prognosis appears **favorable** here.
- **Morbidity:** recurrent TB with associated symptoms and treatment burden; potential long-term pulmonary sequelae (bronchiectasis, fibrosis) as in recurrent TB (extrapolated).

- **Recovery potential:** good per episode with standard anti-TB therapy; **relapse/recurrence** is the main issue.
- **Prognostic factors:** timeliness of diagnosis and treatment, adherence, drug-susceptibility of the infecting strain, and ongoing Mtb exposure. No molecular prognostic biomarker beyond genotype.
- **QoL measures:** not formally assessed.

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12. Treatment

No disease-specific (TNF-replacement) therapy exists. Management is **treatment and prevention of tuberculosis** plus consideration of the underlying immune defect.

- **Anti-tuberculous chemotherapy (standard of care; extrapolated from TB guidelines):** the **RIPE regimen** — **rifampin (NCIT:C769), isoniazid (NCIT:C566), pyrazinamide (NCIT:C1029), ethambutol (NCIT:C61785)** — typically 2 months intensive + 4 months continuation, adjusted for drug susceptibility. Relapses re-treated per susceptibility.
- **Secondary prophylaxis:** consider latent-TB/secondary preventive therapy (isoniazid or rifamycin-based) given recurrence risk; individualized.
- **Avoid TNF inhibitors** in these patients (would compound the defect) — conversely, this disease validates pre-anti-TNF TB screening in the general population.
- **Conceptual/targeted therapy (not yet applied):** because the respiratory-burst defect is **rescued by exogenous TNF** in vitro, **recombinant TNF or TNFR1 agonism** is a mechanistic (but risky/unproven) concept; likewise **recombinant IFN-γ** benefits IFN-γ-circuit TB but rationale is weaker here since the IFN-γ axis is intact. **No gene/cell therapy, RNA therapy, or clinical trials** exist for IMD127.
- **Pharmacogenomics:** standard TB-drug PGx applies (e.g., NAT2 acetylator status and isoniazid toxicity) but is not IMD127-specific.

NCIT intervention terms: Antitubercular Therapy (**NCIT:C15615**/antibiotic therapy), Rifampin (C769), Isoniazid (C566), Pyrazinamide (C1029), Ethambutol (C61785).

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13. Prevention

- **Primary prevention:** avoid/limit Mtb exposure; **BCG is tolerated** and may be given (no BCG-osis), though its protective efficacy in this specific defect is unknown. Household TB source control.
- **Secondary prevention: latent-TB screening (IGRA/TST) and preventive therapy** in genotype-positive relatives; early symptom-triggered evaluation.
- **Tertiary prevention:** adherence support, monitoring for recurrence and pulmonary sequelae.
- **Genetic counseling:** autosomal recessive — 25% recurrence risk per pregnancy for carrier couples; cascade testing; the consanguineous context warrants counseling. Prenatal/ preimplantation testing feasible for the known familial variant.
- **Public-health note:** IMD127 exemplifies why **TB screening is mandated before anti-TNF biologic therapy** in the general population (P 29459143).

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14. Other Species / Natural Disease

- **Taxonomy of affected host:** *Homo sapiens* (NCBI Taxon **9606**).
- **Orthologous gene:** mouse **Tnf** (NCBI Gene **21926**; MGI:104798); rat **Tnf** (NCBI Gene 24835). Highly conserved TNF-superfamily cytokine.
- **Natural animal disease:** No described naturally occurring inherited-TNF-deficiency disease in companion animals/wildlife (OMIA has no corresponding entry). TNF's anti-mycobacterial role is conserved across mammals (mouse KO data).
- **Comparative pathology:** Mouse Tnf/Tnfr1 loss produces **more severe, lethal** mycobacterial disease (failed granulomas, death from attenuated BCG/M. avium) than the narrow human TB-only phenotype — an important species difference (P 10878503 P 11433391 P 10861087).

- **Zoonotic/cross-species transmission:** not applicable (host genetic trait, not a transmissible disease).

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15. Model Organisms

- **Cellular / iPSC (in vitro) — most disease-relevant:** patient-derived and gene-edited **TNF-KO and TNFR1-KO iPSC-derived, GM-CSF-matured macrophages** and **alveolar-macrophage-like (AML) cells**; TNF-blocker-treated healthy MDMs/AML and ex vivo lung macrophages. **Recapitulate** the respiratory-burst defect and demonstrate **TNF rescue** (P 39198650). Cellosaurus/ATCC-type resources apply for iPSC lines.
- **Mouse (mammalian, Mus musculus, Taxon 10090):**
- Available models: **Tnf−/−**, **Tnf/Lta double-KO** (± LTα transgene), **Tnfrsf1a (TNFR1/TNFRp55)−/−**, **Tnfrsf1b (TNFR2)−/−**; conditional/humanized TNF strains via MGI/IMSR/MMRRC.
- **Phenotype recapitulation:** reproduce mycobacterial susceptibility and establish **TNFR1 (not TNFR2)** as the essential receptor (P 10878503); model granuloma biology.
- **Limitations:** phenotype is **broader and lethal** (susceptibility to BCG/M. avium, granuloma necrosis/disintegration, death) — does **not** capture the human specificity for virulent Mtb with sparing of BCG and other microbes; also TNF's roles in murine lymphoid architecture differ.
- **Applications:** dissecting TNF–TNFR1 → respiratory-burst axis; testing TNF/IFN-γ rescue; granuloma dynamics; anti-TNF safety modeling.
- **Resources:** MGI (Tnf MGI:104798), IMSR, MMRRC; Cellosaurus for iPSC/macrophage lines.

Ontology term appendix (for KB population)

- **Disease:** OMIM:620977; MONDO → OMIM:620977 (verify term).

- **Gene/protein:** HGNC:11892 (TNF); NCBI Gene 7124; UniProt P01375; Ensembl ENSG00000232810.
- **HPO:** HP:0032262 (Pulmonary tuberculosis), HP:0032251 (Tuberculosis), HP:0002205 (Recurrent respiratory infections), HP:0002718 (Recurrent bacterial infections).
- **GO (BP):** GO:0045730 (respiratory burst), GO:0006801 (superoxide metabolic process), GO:0071356 (cellular response to TNF), GO:0045087 (innate immune response), GO:0032496 (response to LPS).
- **GO (CC):** GO:0045335 (phagosome), GO:0005615 (extracellular space), GO:0005886 (plasma membrane).
- **CL:** CL:0000583 (alveolar macrophage), CL:0000235 (macrophage), CL:0001054 (CD14+ monocyte).
- **UBERON:** UBERON:0002048 (lung), UBERON:0002299 (alveolus), UBERON:0001004 (respiratory system), UBERON:0002405 (immune system).
- **CHEBI:** CHEBI:26523 (reactive oxygen species), CHEBI:18421 (superoxide).
- **NCIT (treatment):** C769 (rifampin), C566 (isoniazid), C1029 (pyrazinamide), C61785 (ethambutol).
- **NCBI Taxon:** 9606 (human), 10090 (mouse), 1773 (M. tuberculosis), 1639 (L. monocytogenes).

Supported vs. refuted hypotheses

- **Supported:** (1) Biallelic TNF LOF causes isolated susceptibility to pulmonary TB (autosomal recessive). (2) TNF is required for the macrophage respiratory burst that kills Mtb; rescued by TNF. (3) TNFR1 is the essential receptor (mouse + iPSC). (4) Anti-TNF drugs phenocopy the disease (2–5× TB risk). (5) TNF deficiency is mechanistically distinct from IFN-γ-circuit MSMD.
- **Refuted / not supported:** TNF is **not** broadly required for human host defense or systemic inflammation (patients otherwise healthy); the disorder does **not** cause BCG-osis or broad infection susceptibility (contrast with mouse KO and CGD).

Limitations

Single family (n=2); no prevalence, penetrance, long-term survival, or QoL data; treatment is extrapolated from general TB guidelines; no human omics datasets; mouse models over-represent severity. Findings should be revisited as additional families are identified.

Key references (PMID)

- 39198650 — Arias et al., *Nature* 2024 (primary description of IMD127 / inherited TNF deficiency).
- 32055999 — Boisson-Dupuis, monogenic basis of human TB.
- 35985287 — Casanova & Abel, rare-to-common infection genetics.
- 38025345 — Errami & Bousfiha, MSMD state of the art.
- 32602053 — Mahdaviani et al., MSMD cohort.
- 10878503 — Jacobs et al., TNFR1 vs TNFR2 in mycobacterial immunity (mouse).
- 11433391 — Bopst et al., TNF/LTα and BCG (mouse).
- 10861087 — Ehlers et al., TNFRp55 and M. avium granuloma disintegration (mouse).
- 29459143 — Baddley et al., ESGICH anti-TNF safety/TB.
- 42226371 — Alves et al., umbrella review of TB risk with biologics/JAKi.
